Connector

J1 Conn_01x02_Pin: VIN, GND

J2 Conn_01x05_Pin: 3.3V_EXT, 5V_EXT, IO_0_IN, IO_0_OUT

J3 Conn_01x05_Pin: IO_1_IN, IO_1_OUT

J4 Conn_01x05_Pin: IO_2_IN, IO_2_OUT

J5 Conn_01x05_Pin: IO_3_IN, IO_3_OUT

J6 Conn_01x05_Pin: IO_4_IN, IO_4_OUT

GND

RJ11 J7: CAN0_L, CAN0_H

J9 Conn_01x02_Pin

J8 Screw_Terminal_01x04: RELAY0_COM, RELAY0_NO, RELAY1_COM, RELAY1_NO

Input

R3, R10, R6, R12, R8: 27K C22967

R4, R9, R5, R11, R7: 4K7 0.25W C17936

IO_0 → IO_0_IN; IO_3 → IO_3_IN; IO_1 → IO_1_IN; IO_4 → IO_4_IN; IO_2 → IO_2_IN

Protection

D1, D2, D3: BAT54C C37704

RP Pico

D11 B5819W C8598: 5V_INT

SW1 SW_Push C318884

A1 RaspberryPi_Pico

GPIO0 RUN; GPIO1 3V3_EN; GPIO2; GPIO3; GPIO4 CAN_ACT_LED; GPIO5; GPIO6; GPIO7 MCP_INT; GPIO8 SPI_MISO; GPIO9 SPI_CS; GPIO10 SPI_SCK; GPIO11 SPI_MOSI; GPIO12 STATUS_LED; GPIO13 RELAY0_OUT; GPIO14 RELAY1_OUT; GPIO15 IO_4_OUT (R17 470R C23179)

GPIO16 IO_3_OUT; GPIO17 IO_2_OUT; GPIO18 IO_4; GPIO19 IO_3; GPIO20 IO_2; GPIO21 IO_1_OUT; GPIO22 IO_0_OUT (C23179 R16, R15, R13, R14; 470R)

ADC_VREF; GPIO26_ADC0 IO_1; GPIO27_ADC1 IO_0; GPIO28_ADC2 POWER_ADC; AGND

VSYS, VBUS, 3V3

5V_INT 5V_EXT; 3.3V_INT 3.3V_EXT

LED

V_FUSED R19 10K C17902 D5 LED C2286; 5V_INT R20 15K C22809 D6 LED C2296; 3.3V_INT R21 390R C23151 D7 LED C2290

3.3V_INT R18 4K7 C23162 D4 LED C2297 STATUS_LED

3.3V_INT R30 4K7 C23162 D9 LED C2286 CAN_ACT_LED

Relay

V_FUSED; 24V 16.7mA; SM4007 D13 C64898; K1 G5Q-EL3-DC24V; RELAY0_NO, RELAY0_COM

U5 LTV-217 C115450; RELAY0_OUT 390R R24 C23151; 5V_INT R26 2.2K C4190; Q1 AO3400A C20917; 10K R25 C17414

GND

V_FUSED; SM4007 D14 C64898; K2 G5Q-EL3-DC24V; RELAY1_NO, RELAY1_COM

U6 LTV-217 C115450; RELAY1_OUT 390R R27 C23151; 5V_INT R28 2.2K C4190; Q2 AO3400A C20917; 10K R29 C17414

GND

CAN

3.3V_INT C3 100nF C14663; C4 100nF C14663

U2 C626758 MCP2518FD-E/SL: SDI, SDO, CS, SCK, OSC2, OSC1, VDD, VSS, INT0/GPIO0/XSTBY, INT1/GPIO1, RXCAN, TXCAN, CLKO/SOF, INT

SPI_MOSI, SPI_MISO, SPI_CS, SPI_SCK; CAN_RX, CAN_TX, MCP_INT

Y2 YSO110TR C5203551

3.3V_INT C2 100nF C14663; 5V_INT C1 100nF C14663

U3 MCP2542FD-E/SN C220598: TXD, RXD, STBY, VIO, VDD, VSS, CANH, CANL

CAN_TX, CAN_RX

JP1, JP2; R1 60R C23090; R2 60R C23090; C8 4.7nF C53987

CAN0_H, CAN0_L

D12 ESD2CAN24DBZRQ1

Power

V_FUSED; F1 Fuse 1A; D8 SS34 C8678; VIN

C18 10uF C440198; C15 10uF C440198; C14 10uF C440198; C13 10uF C440198; C10 10uF C440198; C9 100nF C14663

U7 XL1509-5.0 C61063: VIN, EN, GND, OUT, FB

L1 68uh C5189964; D10 SS34 C8678; C16 100uF C15008; C17 100uF C15008; C19 10uF C7171

5V_INT

U4 AMS1117-3.3 C6186: VI, GND, VO

5V_INT; C11 100nF C14663; C6 10uF C15850; C7 10uF C7171; C12 100nF C14663; 3.3V_INT

V_FUSED; POWER_ADC; R23 47K C25819; R22 4K7 C23162

GND

H1 MountingHole; H2 MountingHole; H4 MountingHole; H3 MountingHole

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Size: A3 | Date: | **Rev:**
KiCad E.D.A. 9.0.0 | Id: 1/1